AP-E6: The Simply Connected $Sp(4)$ Candidate
Partial Euclidean Controls, Gauge Bordism, Target-Space Eta No-Go, and Heavy-Threshold Audit

Route F independent research note

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Abstract

We complete a sharply delimited audit of the proposed simply connected deep group $Sp(4)_{\mathrm{phys}} = USp(4) = Spin(5) = Sp(2)_{\mathrm{math}}$. First, we give a Euclidean action ansatz and verify two limited fermionic controls for two Dirac $\mathbf{5}$'s: gamma-five reality/positivity and three Pauli–Villars moment identities. This is *not* a complete Euclidean regulator. In particular, the regulator-field statistics and full mass interpolation, APS boundary data, finite local-counterterm scheme, and regulated gauge–scalar–ghost measure have not been specified.

Second, an explicit Atiyah–Hirzebruch spectral-sequence computation gives

$$\boxed{\Omega_5^{\mathrm{Spin}}(BSp(2)) \simeq \mathbb{Z}_2.}$$

Its generator is an embedded unit $Sp(1)$ instanton on S^4, times the periodic spin circle, equivalently the mapping torus of the nontrivial element of $\pi_4(Sp(2))$. Restriction to this $Sp(1)$ gives mod-two indices $\nu_{\mathbf{4}} = 1$, $\nu_{\mathbf{5}} = 0$, and $\nu_{\mathbf{10}} = 0$. Hence the displayed two Dirac $\mathbf{5}$'s define the trivial character of the nonzero bordism class. The dynamical $Sp(4)$ gauge-bordism anomaly gate is therefore closed for this field list.

Third, this result does *not* select the two target-space signs on $G_3 = S^1_{\mathrm{R}} \times S^3$ and $G_2 = T^2_{\mathrm{RR}} \times S^2$. Those signs are characters of $\widetilde{\Omega}_4^{\mathrm{Spin}}(S^3 \times S^2)$, and require a mass map from the sigma-model target into the regulated Fredholm family. The current $Sp(4)$ action specifies neither the composite pion lift nor the full s-dependent Dirac–Yukawa/PV family. We prove underdetermination by enumerating the independent torsion invertible sectors/counterterms that flip either character while leaving the gauge character and de Rham curvature fixed. This enumeration is not a microscopic realization of those sectors.

Finally, threshold bookkeeping produces a stronger obstruction. The $\mathbf{5}$ branches as $\mathbf{2}_{+1} \oplus \mathbf{2}_{-1} \oplus \mathbf{1}_0$. Uniform flavour action on the complete irrep gives $2(+1) + 2(-1) + 0 = 0$. Under the additional hypothesis that the chiral flavour symmetry acts uniformly and exactly on the complete $\mathbf{5}$, the light/heavy charge traces are conditionally $+2$ and -2. This is an algebraic obstruction, not an evaluated heavy determinant. The physical $B = +1$ infrared orientation $k_{\mathrm{IR}} = +2$ is derived, but it can survive only if the chiral flavour symmetry is emergent below the breaking threshold or if new UV topological data are added. Neither option is yet a closed first-principles completion. No Route-E portal is authorized.

Contents

1 Scope, notation, and exact claim boundary

Throughout this note "$Sp(4)$" means the physics group with a four-dimensional defining representation,

$$G = Sp(4)_{\text{phys}} = USp(4) = Spin(5) = Sp(2)_{\text{math}}. \tag{1}$$

This convention is essential when reading bordism tables. The candidate field content is

$$\Psi_i \in \mathbf{5} \text{ Dirac}, \quad i = 1, 2; \qquad \Phi_I \in \mathbf{4} \text{ complex}, \quad I = 1, 2; \qquad \Sigma \in \mathbf{10} \text{ real}. \tag{2}$$

The copy index I carries the desired global $SU(2)_\phi$; it is not a gauge weight. The standard branch through $Spin(4) = SU(2)_c \times SU(2)_H$ is

$$\mathbf{4} \longrightarrow (\mathbf{2}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{2}), \tag{3}$$
$$\mathbf{5} \longrightarrow (\mathbf{2}, \mathbf{2}) \oplus (\mathbf{1}, \mathbf{1}), \tag{4}$$
$$\mathbf{10} \longrightarrow (\mathbf{3}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{3}) \oplus (\mathbf{2}, \mathbf{2}). \tag{5}$$

After $SU(2)_H \to U(1)_H$, with doublet weights normalized to $q = \pm 1$,

$$\mathbf{4} \to \mathbf{2}_0 \oplus \mathbf{1}_{+1} \oplus \mathbf{1}_{-1}, \tag{6}$$
$$\mathbf{5} \to \mathbf{2}_{+1} \oplus \mathbf{2}_{-1} \oplus \mathbf{1}_0, \tag{7}$$
$$\mathbf{10} \to \mathbf{1}_{+2} \oplus \mathbf{2}_{+1} \oplus \mathbf{3}_0 \oplus \mathbf{1}_0 \oplus \mathbf{2}_{-1} \oplus \mathbf{1}_{-2}. \tag{8}$$

These identities can be obtained from $\mathbf{5} = \wedge^2 \mathbf{4} - \mathbf{1}$ and $\mathbf{10} = \mathrm{Sym}^2\, \mathbf{4}$ [1].

Four logically different questions will not be conflated:

(i) whether the Euclidean $Sp(4)$ path integral has been specified;

(ii) whether its *dynamical gauge* determinant line has a global anomaly, classified here by $\Omega_5^{\text{Spin}}(BSp(2))$;

(iii) which spin-torsion characters the induced *target-space* functional assigns to G_3 and G_2;

(iv) whether the heavy threshold preserves the desired integer mixed-WZW coefficient and the $k = +2$ orientation.

Only (ii) will close. Question (i) has only partial fermion controls, (iii) is proved underdetermined, and (iv) has a conditional uniform-flavour trace obstruction but no APS determinant-ratio computation. This separation follows the APS/Dai–Freed treatment of determinant phases [2, 3, 4].

2 Partial Euclidean fermion-regulator controls

2.1 Bundles, hermiticity, and the bosonic action

Let M_4 be a closed oriented Riemannian spin manifold and $P \to M_4$ a principal G-bundle. Write $E_R = P \times_G R$, choose Hermitian Euclidean gamma matrices, and use an anti-Hermitian gauge connection A. If T_R^a are anti-Hermitian generators, define the Hermitian adjoint-Higgs endomorphism

$$\widehat{\Sigma}_R := -\mathrm{i}\Sigma^a T_R^a, \qquad \widehat{\Sigma}_R^\dagger = \widehat{\Sigma}_R. \tag{9}$$

The gauge-fixed bosonic action may be chosen as

$$S_{E,\mathrm{bos}} = \int_{M_4} \mathrm{d}^4x\sqrt{g}\,\mathcal{L}_{\mathrm{bos}} + S_{\mathrm{gf}} + S_{\mathrm{gh}}, \tag{10}$$

$$\begin{aligned}\mathcal{L}_{\mathrm{bos}} &= \frac{1}{2g_G^2}\operatorname{tr}(F_{\mu\nu}^\dagger F^{\mu\nu}) + \frac{1}{2}(D_\mu\Sigma)^a(D^\mu\Sigma)^a \\ &\quad + \sum_{I=1}^{2}(D_\mu\Phi_I)^\dagger D^\mu\Phi_I + V(\Sigma,\Phi),\end{aligned} \tag{11}$$

$$\begin{aligned}V &= \frac{\lambda_\Sigma}{4}\left(\Sigma^a\Sigma^a - v_\Sigma^2\right)^2 + m_\Phi^2\sum_I \Phi_I^\dagger\Phi_I + \kappa\sum_I \Phi_I^\dagger\widehat{\Sigma}_{\mathbf{4}}\Phi_I \\ &\quad + \lambda_1\Big(\sum_I \Phi_I^\dagger\Phi_I\Big)^2 + \lambda_2\sum_{I,J}|\Phi_I^\dagger\Phi_J|^2.\end{aligned} \tag{12}$$

Here $\lambda_\Sigma, \lambda_1 > 0$ and $\lambda_1 + \lambda_2 > 0$ give a simple bounded quartic domain. The copy-symmetric coefficients preserve $SU(2)_\phi$. A background R_ξ gauge fixing and its Faddeev–Popov determinant define $S_{\mathrm{gf}} + S_{\mathrm{gh}}$ formally; no regulator for the gauge–scalar–ghost complex is supplied here. This omission does not affect the separate bordism computation below, but it keeps the full Euclidean-regulator gate open.

Choose an adjoint vacuum

$$\langle\widehat{\Sigma}\rangle = VH, \qquad H = -2\mathrm{i}T_H^3, \tag{13}$$

where T_H^3 is anti-Hermitian. Thus H is Hermitian and has the real weights displayed in Eqs. (6)–(8). Because $Spin(4) \hookrightarrow Spin(5)$ is injective, its central element $(-\mathbf{1}_c, -\mathbf{1}_H)$ maps to the nontrivial centre of $Spin(5)$, not to the identity. The unbroken group is consequently the direct product $SU(2)_c \times U(1)_H$, rather than its diagonal $\mathbb{Z}_2$ quotient.

2.2 Dirac–Yukawa operator and threshold masses

For each flavour let

$$\mathcal{K}_{A,\Sigma} := \gamma^\mu(\nabla_\mu + A_\mu^{(\mathbf{5})}) + m\mathbf{1}_{\mathbf{5}} + y\widehat{\Sigma}_{\mathbf{5}}. \tag{14}$$

The fermion action is

$$S_{E,\mathrm{f}} = \sum_{i=1}^{2}\int_{M_4}\mathrm{d}^4x\sqrt{g}\,\overline{\Psi}_i\mathcal{K}_{A,\Sigma}\Psi_i. \tag{15}$$

On the three branches of $\mathbf{5}$,

$$M_+ = m + yV, \qquad M_- = m - yV, \qquad M_0 = m. \tag{16}$$

At the tree-level light-point $m = -yV$, and with $yV > 0$,

$$(M_+, M_-, M_0) = (0, -2yV, -yV). \tag{17}$$

For determinant statements we keep a positive infrared mass $M_+ = \mu > 0$ and take $\mu \downarrow 0$ only after all phases and zero modes have been recorded.

The operator obeys

$$\mathcal{K}_{A,\Sigma}^{\dagger} = \gamma_5 \mathcal{K}_{A,\Sigma} \gamma_5. \tag{18}$$

Thus its nonreal eigenvalues occur in complex-conjugate pairs and $\det \mathcal{K} \in \mathbb{R}$. For the two identical flavours,

$$Z_{\Psi,\mu} = \left(\det \mathcal{K}_{A,\Sigma;\mu}\right)^2 \geq 0 \tag{19}$$

away from the declared zero locus. This is a two-flavour positivity statement, not a claim that an arbitrary chiral mass family has positive determinant.

2.3 Pauli–Villars amplitude and eta phase

It is useful to separate the ultraviolet amplitude subtraction from the global phase. Put

$$c_a = (1, -3, 3, -1), \qquad \Lambda_a^2 = a\Lambda^2, \qquad a = 0, 1, 2, 3, \tag{20}$$

The finite-difference identities

$$\sum_a c_a = \sum_a c_a a = \sum_a c_a a^2 = 0 \tag{21}$$

remove the first three heat-kernel power moments in

$$\log |\det \mathcal{K}|_{\mathrm{PV}} = \frac{1}{2} \sum_{a=0}^{3} c_a \log \det \left(\mathcal{K}^{\dagger}\mathcal{K} + \Lambda_a^2\right). \tag{22}$$

The remaining logarithmic divergences would have to be removed by a declared finite gauge-invariant local-counterterm scheme. Such a scheme is not supplied in this note.

For the phase, a completed construction would choose a five-dimensional interpolation Y_5, self-adjoint operators $D_{Y,R,a}$, compatible APS boundary conditions, regulator statistics, and PV reference signs $\sigma_a \in \{\pm 1\}$. Its torsion datum would have the schematic form

$$\tau_R(Y_5; \sigma) = \exp\left[2\pi \mathrm{i} \sum_a c_a \sigma_a \overline{\eta}(D_{Y,R,a})\right], \qquad \overline{\eta} = \frac{\eta + \dim \ker D}{2}. \tag{23}$$

Once all boundary and interpolation data are supplied, APS gluing can make the torsion part a bordism character. Here Eq. (23) is only a template: neither the full $\mathcal{K}(t), \Lambda_a(t)$ mass interpolation nor the APS boundary operator/domain, local counterterms, and gauge–scalar–ghost regulator are defined.

Proposition 2.1 (Partial fermion-regulator controls)**.** *Equations* (18)–(21) *establish gamma-five reality, two-flavour positivity away from the declared zero locus, and three PV moment identities. They do not define the complete functional measure or its phase. Consequently*

$$\texttt{fermion\_pv\_gamma5\_controls\_verified=true}, \qquad \texttt{sp4\_euclidean\_regulator\_complete=false}. \tag{24}$$

3 The exact gauge-bordism group

3.1 AHSS computation

The integral cohomology ring is

$$H^*(BSp(2); \mathbb{Z}) = \mathbb{Z}[q_1, q_2], \qquad |q_1| = 4, \quad |q_2| = 8. \tag{25}$$

Consequently, in degrees below eight,

$$H_p(BSp(2); \mathbb{Z}) = \begin{cases} \mathbb{Z}, & p = 0, 4, \\ 0, & 0 < p < 8, \ p \neq 4, \end{cases} \tag{26}$$

and $H_4(BSp(2);\mathbb{Z}_2)=\mathbb{Z}_2$. The homological AHSS is

$$E^2_{p,q}=H_p(BSp(2);\Omega^{\mathrm{Spin}}_q)\Longrightarrow\Omega^{\mathrm{Spin}}_{p+q}(BSp(2)),\qquad d_r:E^r_{p,q}\to E^r_{p-r,q+r-1}. \tag{27}$$

Use

$$\Omega^{\mathrm{Spin}}_0=\mathbb{Z},\quad\Omega^{\mathrm{Spin}}_1=\mathbb{Z}_2,\quad\Omega^{\mathrm{Spin}}_2=\mathbb{Z}_2,\quad\Omega^{\mathrm{Spin}}_3=0,\quad\Omega^{\mathrm{Spin}}_4=\mathbb{Z},\quad\Omega^{\mathrm{Spin}}_5=0. \tag{28}$$

On the total-degree-five diagonal the only nonzero term is

$$E^2_{4,1}=H_4(BSp(2);\mathbb{Z}_2)=\mathbb{Z}_2. \tag{29}$$

The outgoing maps are

$$\begin{aligned} d_2&:E^2_{4,1}\longrightarrow E^2_{2,2}=0,\\ d_3&:E^3_{4,1}\longrightarrow E^3_{1,3}=0,\\ d_4&:E^4_{4,1}\longrightarrow E^4_{0,4}=\mathbb{Z}. \end{aligned}\tag{30}$$

The last homomorphism vanishes because there is no nonzero group homomorphism $\mathbb{Z}_2\to\mathbb{Z}$. There is no incoming d_2, since $E^2_{6,0}=H_6(BSp(2);\mathbb{Z})=0$, and for $r\geq 3$ an incoming differential would start at negative coefficient degree. Hence the class survives to E^∞, and there is no extension problem because it is the only filtration quotient.

Theorem 3.1 (Low-degree $Sp(4)$ gauge bordism)**.** *For the simply connected group in Eq.* (1)*,*

$$\boxed{\Omega^{\mathrm{Spin}}_5(BSp(2))\simeq\mathbb{Z}_2.} \tag{31}$$

This agrees with the direct $Sp(n)$ analysis of Saito and Tachikawa [5]; recent instanton zero-mode analysis is also given by Jia and Yi [6].

3.2 Geometric generator and representation character

Let $P_1\to S^4$ be the unit instanton bundle for an embedded $Sp(1)=SU(2)_c\subset Sp(2)$. A geometric generator is

$$Y_5=(S^4,P_1)\times S^1_{\mathrm{R}}. \tag{32}$$

It is bordant to the mapping torus obtained by gluing the ends of $S^4\times[0,1]$ with the nontrivial element of $\pi_4(Sp(2))=\mathbb{Z}_2$, and also to the S^5 clutching construction. For a four-dimensional left Weyl fermion in representation R, product factorization gives

$$\tau_R(Y_5)=(-1)^{\mathrm{ind}_{S^4}D_R}. \tag{33}$$

Restrict Eqs. (3)–(5) to the instanton $SU(2)_c$. The required decompositions and instanton indices are

$$\mathbf{4}\downarrow SU(2)_c=\mathbf{2}\oplus\mathbf{1}\oplus\mathbf{1},\qquad \mathrm{ind}\,D_{\mathbf{4}}=1, \tag{34}$$

$$\mathbf{5}\downarrow SU(2)_c=2\,\mathbf{2}\oplus\mathbf{1},\qquad \mathrm{ind}\,D_{\mathbf{5}}=2, \tag{35}$$

$$\mathbf{10}\downarrow SU(2)_c=\mathbf{3}\oplus 2\,\mathbf{2}\oplus 3\,\mathbf{1},\qquad \mathrm{ind}\,D_{\mathbf{10}}=4+2=6. \tag{36}$$

Here $2T_{SU(2)}(\mathbf{2})=1$ and $2T_{SU(2)}(\mathbf{3})=4$. Equivalently, in the convention $T_{Sp(2)}(\mathbf{4})=1/2$,

$$2T(\mathbf{4})=1,\qquad 2T(\mathbf{5})=2,\qquad 2T(\mathbf{10})=6. \tag{37}$$

Therefore

$$\nu_{\mathbf{4}}=1,\qquad\nu_{\mathbf{5}}=0,\qquad\nu_{\mathbf{10}}=0\quad\text{in }\mathbb{Z}_2. \tag{38}$$

The fundamental **4** detects the Witten class [7, 8]; the vector **5** and adjoint **10** do not.

A Dirac **5** is two left-Weyl **5**'s after charge conjugation. For two Dirac flavours the exponent is

$$\nu_{2\mathrm{D}\,\mathbf{5}}=2\text{ flavours}\times 2\text{ Weyl/Dirac}\times\nu_{\mathbf{5}}=4\times 0=0\pmod 2. \tag{39}$$

Scalars do not contribute a fermionic Dai–Freed character.

Corollary 3.2 (Gauge-bordism gate closes)**.** *The complete dynamical $Sp(4)$ gauge determinant for the displayed fermion list evaluates to $+1$ on the generator of $\Omega_5^{\mathrm{Spin}}(BSp(2))$. Thus*

$$\texttt{sp4\_gauge\_bordism\_character\_trivial=true}. \tag{40}$$

This conclusion is stronger than merely observing that the fields are vectorlike: the nonzero bordism group and its representation character have both been computed.

4 Why the two target mapping-torus phases remain undetermined

4.1 Different domains for the two characters

Put $X = S^3 \times S^2$. The target-space spin refinement is a character of

$$\widetilde{\Omega}_4^{\mathrm{Spin}}(X) = \mathbb{Z}_2 \oplus \mathbb{Z}_2, \tag{41}$$

with generators

$$G_3 = (S^1_{\mathrm{R}} \times S^3, \mathrm{pr}_{S^3}, \mathrm{pt}), \tag{42}$$

$$G_2 = (T^2_{\mathrm{RR}} \times S^2, \mathrm{pt}, \mathrm{pr}_{S^2}). \tag{43}$$

The most general torsion factor is

$$Z_{\mathrm{tor}}(M, g, s) = (-1)^{\epsilon_3 \nu_3(M,g) + \epsilon_2 \nu_2(M,s)}, \qquad (\epsilon_3, \epsilon_2) \in \mathbb{Z}_2^2. \tag{44}$$

Here the invariants have a concrete mod-two definition. For regular values $p_3 \in S^3$ and $p_2 \in S^2$, give $L_g = g^{-1}(p_3)$ and $\Sigma_s = s^{-1}(p_2)$ their induced spin structures and set

$$\nu_3(M, g) = \dim\ker D_{L_g} \pmod 2 \in \Omega_1^{\mathrm{Spin}} = \mathbb{Z}_2, \qquad \nu_2(M, s) = \mathrm{Arf}(\Sigma_s) \in \Omega_2^{\mathrm{Spin}} = \mathbb{Z}_2. \tag{45}$$

Their exponentials are abstract invertible spin-TQFT characters. They may be stacked as torsion counterterms/sectors, but Eq. (44) does not by itself construct a massive microscopic regulator realizing any chosen pair.

By contrast, Eq. (31) classifies gauge bundles on five-manifolds. There is no canonical map

$$\widetilde{\Omega}_4^{\mathrm{Spin}}(S^3 \times S^2) \longrightarrow \Omega_5^{\mathrm{Spin}}(BSp(2)). \tag{46}$$

To produce a determinant phase one must specify a regulated mass-family lift

$$\mathfrak{L}:\ (g, s) \longmapsto [A(g, s), \Sigma(g, s), \Phi(g, s), \mathcal{M}(g, s), \{c_a, \Lambda_a, \sigma_a\}] \in \mathcal{F}_{\mathrm{Fred}}/\mathcal{G}, \tag{47}$$

and then pull back the determinant line and its Dai–Freed connection. The action in Section 2 only contains the elementary A, Σ, Φ. It does not specify the composite pion g-dependent chiral mass, nor a complete s-dependent fermion Yukawa operator, nor the PV continuation along either generator.

Theorem 4.1 (Target eta underdetermination)**.** *The computation $\Omega_5^{\mathrm{Spin}}(BSp(2)) = \mathbb{Z}_2$ and the trivial gauge character of the two Dirac* **5***'s do not determine (ϵ_3, ϵ_2). With the current field/action data, neither $\tau(G_3)$ nor $\tau(G_2)$ is a defined microscopic number.*

Proof. The two putative characters have different bordism domains, as shown by Eqs. (41) and (31). A comparison requires the extra map $\mathfrak{L}$ in Eq. (47). It is absent. Independently, the four characters in Eq. (44) are distinguished by the two mod-two indices in Eq. (45). Stacking either abstract invertible torsion sector has zero de Rham curvature and does not alter the $Sp(4)$ gauge character or perturbative anomaly polynomial, but it flips one target character. Because the microscopic theory has not fixed these invertible sectors/counterterms, all four character assignments remain compatible with the supplied local and gauge-bordism data. This is an ambiguity statement, not a construction of four heavy determinants. □

The explicit negative-control table is

invertible sector/counterterm	(ϵ_3, ϵ_2)	$Z(G_3)$	$Z(G_2)$
reference	$(0,0)$	$+1$	$+1$
$I_3 = (-1)^{\nu_3}$	$(1,0)$	-1	$+1$
$I_2 = (-1)^{\nu_2}$	$(0,1)$	$+1$	-1
$I_3 I_2$	$(1,1)$	-1	-1

This table enumerates the character group and proves non-uniqueness of the present data. It is neither a menu from which the desired answer may be selected nor evidence that any row has been generated by the displayed fermion determinant.

4.2 Conditional spectral controls

Define a conditional no-defect reference $\mathcal{C}_0$ on the retained branch by

$$\mathcal{K}_{\mathcal{C}_0}(g, s) = \not{D}_{A(s)} + \mu\big(P_L g + P_R g^\dagger\big), \qquad \mu > 0, \tag{48}$$

with the $q = -1, 0$ branch masses kept constant, a formally positive PV reference orientation, and no I_3 or I_2 counterterm. Here $A(s)$ is the pullback Hopf connection seen by the charge-one light branch. This is a formal *conditional low-energy* parity control; it is not the missing all-field deep-$Sp(4)$ lift in Eq. (47).

Two useful parity controls then follow. In the standard light two-colour chiral mass family, the S^3 discrete-WZW parity is

$$N_c \pmod 2 = 2 \pmod 2 = 0. \tag{49}$$

For a unit Hopf line over the S^2 factor, the number of light charged zero-mode copies is

$$N_c N_f |c_1| = 2 \cdot 2 \cdot 1 = 4 = 0 \pmod 2. \tag{50}$$

Thus the formal parity count in $\mathcal{C}_0$ would return $(+1, +1)$. It is not an evaluated microscopic eta invariant. Stacking I_3 or I_2 changes the character without changing the stated light spectra. Equations (49) and (50) therefore cannot be promoted to a unique microscopic selection.

Fail-closed gate 4.2 (Mapping-torus pair)**.** Until Eq. (47) is supplied and the two APS problems are evaluated with the same PV convention,

$$\texttt{target\_mapping\_torus\_eta\_pair\_selected=false}. \tag{51}$$

5 Heavy thresholds and the fate of $k = +2$

5.1 Weight, multiplicity, mass-sign, and index ledger

For $yV > 0$, $m = -yV$, and the positive light regulator mass μ, the complete one-flavour branch ledger is

branch	$U(1)_H$ charge	$SU(2)_c$	multiplicity	mass	sign
light	$+1$	$\mathbf{2}$	2	μ	$+$
charged heavy	-1	$\mathbf{2}$	2	$-2yV$	$-$
neutral heavy	0	$\mathbf{1}$	1	$-yV$	$-$

There are two identical flavour copies. Mass signs control theta-like threshold terms relative to the PV reference, whereas the mixed flavour coefficient below is a charge trace. These two ledgers must not be identified.

For negative real masses and a positive PV reference, the pure unbroken gauge theta shifts, in units of π, are

$$\frac{\Delta\theta_c}{\pi} = N_f\,[2T_{SU(2)}(\mathbf{2})] = 2\cdot 1 = 2, \tag{52}$$

$$\frac{\Delta\theta_H}{\pi} = N_f\,[\dim \mathbf{2}\, q^2] = 2\cdot 2\cdot 1 = 4. \tag{53}$$

Both exponentiate trivially on integral instanton sectors. The neutral branch has no gauge theta shift. Flipping all PV reference signs changes these integers by even amounts for the two-flavour spectrum. This is only a local pure-gauge theta-periodicity check in the stated normalization. It is *not* an APS eta or determinant-ratio matching calculation. The unbroken-group Dai–Freed ratio, gravitational phase, spectral flow, and boundary zero-mode contribution remain open.

As a second negative control, take $m = +yV$. Then

$$(M_+, M_-, M_0) = (2yV, 0, yV), \tag{54}$$

so the $q=-1$, not the $q=+1$, colour doublet is light. Its infrared orientation is $k=-2$. This shows that the sign of k_{IR} is fixed by the selected branch and generator orientation, not by the positivity of the full two-flavour measure.

5.2 Conditional uniform-flavour charge-trace obstruction

Let $SU(2)_L \times SU(2)_R$ act uniformly on the two flavour copies of a complete $\mathbf{5}$. In the normalization where a retained $\mathbf{2}_{+1}$ contributes $+2$, the left mixed coefficient is

$$\kappa_L^{\mathrm{full}} = \sum_{w\in\mathbf{5}} q_w = 2(+1) + 2(-1) + 1(0) = 0, \qquad \kappa_R^{\mathrm{full}} = -\kappa_L^{\mathrm{full}} = 0. \tag{55}$$

Splitting light and heavy pieces gives

$$\kappa_L^{\mathrm{light}} = +2, \qquad \kappa_L^{\mathrm{heavy}} = -2, \qquad \kappa_L^{\mathrm{light}} + \kappa_L^{\mathrm{heavy}} = 0. \tag{56}$$

These are representation-theory charge traces, not terms extracted from an evaluated regulated determinant. The split acquires threshold meaning only under the uniform exact-flavour and symmetry-preserving matching hypotheses stated next. This equality follows from tracelessness of the embedded semisimple generator. It is independent of the signs in Eq. (17). Indeed, multiplying an $SU(2)$ chiral mass map by -1 sends $g \mapsto -g$, but

$$(-g)^{-1}\mathrm{d}(-g) = g^{-1}\mathrm{d}g, \tag{57}$$

so the normalized S^3 winding form is unchanged. Mass sign affects the separate theta ledger, not the charge-trace identity.

Proposition 5.1 (Conditional uniform-flavour trace obstruction)**.** *Assume the chiral flavour symmetry acts uniformly on a complete $Sp(4)$ $\mathbf{5}$, and the heavy threshold preserves that exact symmetry. Then the complete-irrep charge trace is zero, while its formal light/heavy decomposition is $+2-2$. Hence any symmetry-preserving matching construction would have to account for the heavy -2 rather than simply discard it. This conditional obstruction does not assert that the actual heavy determinant has been calculated.*

Proof. The assumed uniform flavour assignment makes the relevant representation trace additive over the five weights. Equations (55) and (56) are then exact algebraic identities. Promoting them to a statement about determinant phases requires the missing regulated mass interpolation and APS ratio; no such promotion is made here. □

This is the concrete $Sp(4)$ realization of the recent semisimple deep-UV warning: a nonzero mixed WZW/2-group structure requires the relevant flavour and magnetic symmetries to emerge together below the semisimple breaking scale, rather than remain exact throughout the deep UV [9].

The actual operator $m + y\widehat{\Sigma}_{\mathbf{5}}$ makes this distinction unavoidable. A nonzero heavy-branch Dirac mass preserves only the vector flavour subgroup, not an exact independent $SU(2)_L \times SU(2)_R$ acting on the full $\mathbf{5}$. The candidate can therefore evade the hypothesis of the proposition only by treating the desired chiral symmetry as emergent in the tuned light branch. That is a logically consistent escape route, but it no longer supplies a deep-UV anomaly or 2-group matching derivation of $k = +2$; the emergence and threshold functional must be constructed explicitly.

5.3 What remains true about physical orientation

Choose the generator H so the retained colour doublet has $X_q = +1$, and orient the baryon current so a soliton has $B = +1$. The light charged-two-colour sector then has

$$n_{\mathrm{IR}} = N_c X_q = 2, \qquad k_{\mathrm{IR}} = n_{\mathrm{IR}} B = +2. \tag{58}$$

The opposite branch in Eq. (54) gives -2, and CPT sends $B \to -B$. Thus Eq. (58) is a physical, declared infrared orientation. It is not an all-scale matching theorem.

There are three logically possible repairs:

1. make $SU(2)_L \times SU(2)_R$ emergent only after $Sp(4) \to SU(2)_c \times U(1)_H$, while explicitly tracking the symmetry-emergence scale and every threshold;
2. add a specified heavy invertible sector whose integer mixed functional changes the threshold by $+2$, then compute its two target torsion signs with the same regulator;
3. abandon the uniform complete-irrep flavour assignment and construct a different anomaly-consistent UV representation set.

Option 1 is closest to the known deep-UV mechanism, but none has yet been implemented in this candidate.

Fail-closed gate 5.2 (Full heavy-threshold eta matching)**.** Although the pure gauge theta shifts in Eqs. (52)–(53) pass a local periodicity check, the unbroken-group and gravitational APS determinant ratios, target-dependent heavy determinant, and all-scale $+2$ coefficient are not computed. Therefore

$$\begin{gathered}\texttt{local\_pure\_gauge\_theta\_periodicity\_check=true},\\ \texttt{unbroken\_group\_aps\_determinant\_ratio\_computed=false},\\ \texttt{gravitational\_aps\_phase\_computed=false},\\ \texttt{heavy\_threshold\_eta\_matched=false},\\ \texttt{k\_plus\_two\_ir\_orientation\_derived=true},\\ \texttt{k\_plus\_two\_all\_scale\_matched=false}.\end{gathered} \tag{59}$$

6 Mechanical and numerical verification

The companion script performs exact rational/integer checks and small spectral controls. Its independent tests include:

1. the low-degree $BSp(2)$ AHSS diagonal and every possible incoming and outgoing differential;
2. the $SU(2)_c$ restriction of $\mathbf{4}, \mathbf{5}, \mathbf{10}$ and mod-two Dynkin indices $(1, 0, 0)$;
3. the four-Weyl multiplicity of two Dirac $\mathbf{5}$'s;
4. the partial fermionic PV moments and gamma-five controls, together with an explicit check that the full regulator gate remains false;

5. both mass tunings, all charge multiplicities, and the local pure-gauge theta-periodicity shifts $(2, 4)$;
6. the conditional uniform-flavour trace identity $+2 - 2 = 0$, without claiming an APS determinant evaluation;
7. finite-mode gamma-five pairing and reference zero-mode parities for G_3, G_2;
8. the four abstract target invertible characters, their mod-two-index definitions, and all fail-closed booleans.

Passing these checks validates the algebra and the conditional obstruction. It does not convert a false physics gate into a true one.

7 Final theorem and gate ledger

Theorem 7.1 (AP-E6 result)**.** *For the displayed simply connected $Sp(4)$ field list:*

1. *fermionic gamma-five and PV-moment controls are explicit, but the complete Euclidean regulator remains open;*
2. *$\Omega_5^{\mathrm{Spin}}(BSp(2)) = \mathbb{Z}_2$, and the two Dirac $\mathbf{5}$'s have trivial character on its generator;*
3. *the target mapping-torus pair remains underdetermined because the necessary mass-family lift is absent and abstract invertible torsion sectors/counterterms can flip either character;*
4. *the local pure-gauge theta shifts pass a periodicity check, while a uniform complete-irrep flavour hypothesis gives only the conditional charge-trace obstruction $+2 - 2 = 0$; no heavy APS ratio is evaluated;*
5. *the light $B = +1$ sector has the physical orientation $+2$, but no all-scale $+2$ matching, radiative protection, or nonperturbative $B = 1$ phase is established.*

Consequently this work closes the dynamical gauge-bordism subgate, but not the Euclidean-regulator, heavy-threshold, or full $Sp(4)$ lane.

Gate	status	evidence
Fermion PV/gamma-five controls	partial	Eqs. (18)–(21)
Complete Euclidean $Sp(4)$ regulator	open	mass interpolation, APS boundary/local scheme, and bosonic regulators absent
$\Omega_5^{\mathrm{Spin}}(BSp(2))$	closed	AHSS gives the unique surviving $E^2_{4,1} = \mathbb{Z}_2$
Displayed gauge-bordism character	closed	$\nu_{\mathbf{5}} = 0$, hence two Dirac $\mathbf{5}$'s are trivial
Target (ϵ_3, ϵ_2)	open	mass-family lift absent; four invertible characters remain possible
Local pure-gauge theta periodicity	checked	local shifts $(2\pi, 4\pi)$ exponentiate trivially
Unbroken/gravitational APS threshold	open	no regulated determinant ratio, spectral flow, or boundary analysis
Target-dependent heavy threshold	open	uniform exact flavour only conditionally gives the trace $+2 - 2 = 0$
All-scale $k = +2$	open	only the light $B = +1$ orientation is derived

Gate	status	evidence
Radiative protection	open	$m = -yV$ remains an unprotected cancellation
Strong charged $B = 1$ phase	open	no dynamical lattice/continuum soliton proof in this note
Full lane / Route-E portal	blocked	target eta, threshold, naturalness, and dynamics gates remain false

The machine-readable status is

$$\begin{gathered}\texttt{fermion\_pv\_gamma5\_controls\_verified=true},\\ \texttt{sp4\_euclidean\_regulator\_complete=false},\\ \texttt{omega5\_spin\_bsp4\_computed=true},\\ \texttt{target\_mapping\_torus\_eta\_pair\_selected=false},\\ \texttt{local\_pure\_gauge\_theta\_periodicity\_check=true},\\ \texttt{heavy\_threshold\_eta\_matched=false},\\ \texttt{threshold\_radiatively\_protected=false},\\ \texttt{strong\_b1\_phase\_proven=false},\\ \texttt{lane\_closed=false}, \qquad \texttt{physics\_promotion\_allowed=false}.\end{gathered} \tag{60}$$

8 Next discriminating calculations

The following alternatives are ordered by information gain.

1. **Emergent-flavour construction.** Specify the fields that split the full **5** flavour symmetry at the $Sp(4)$ breaking scale, derive the exact light flavour symmetry, and calculate the induced integer threshold functional. This directly tests whether the semisimple no-go assumptions are escaped rather than merely violated.
2. **Common target mass map.** Construct $\mathfrak{L}(g, s)$ in Eq. (47) on both G_3 and G_2, including every heavy branch and PV sign. A single lattice spectral-flow implementation should then compute both eta phases without independent convention changes.
3. **Alternative representation set.** Search for complete $Sp(4)$ representations plus spectators whose exact flavour-weighted charge trace is $+2$, whose $\mathbb{Z}_2$ gauge-bordism character vanishes, and whose scalar sector still realizes the charge-one/exactly-two operator. The trace and mod-two-index constraints can be imposed before any dynamical scan.

No degree-one portal should be built until at least one full lane, including its nonperturbative $B = 1$ gate, closes.

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